

Configuring VLANs & 802.1Q Trunking in Packet Tracer

Standard Operating Procedure & Implementation Guide for Cisco 2960 Switches

Document ID: NET-DOC-VLAN-001 Author: Luigi Cocjin Environment: Cisco Packet Tracer Date: August 25, 2026

1. Network Topology & Overview

This document outlines the step-by-step procedure to configure VLAN segmentation and 802.1Q trunking across two Cisco 2960 Switches (Switch0 and Switch1). Inter-switch traffic is linked via interface **FastEthernet 0/1**, utilizing **VLAN 99** as the native VLAN.

Key Objective: Enable isolated broadcast domains (VLAN 10 and VLAN 20) while allowing management access via Switch Virtual Interfaces (SVIs).

2. IP Addressing & Interface Specification

Device / Host	Interface / Port	IP Address	Subnet Mask	VLAN	Role / Description
Switch0	SVI (Vlan 10)	192.168.10.2	255.255.255.0	10	Switch Management
Switch0	Fa0/1	N/A	N/A	99 (Native)	Trunk Link to Switch1
PC0	Fa0	192.168.10.4	255.255.255.0	10	VLAN 10 Host
PC1	Fa0	192.168.10.5	255.255.255.0	10	VLAN 10 Host
Switch1	SVI (Vlan 10)	192.168.10.3	255.255.255.0	10	Management (VLAN 10)
Switch1	SVI (Vlan 20)	192.168.20.1	255.255.255.0	20	Management (VLAN 20)
Switch1	Fa0/1	N/A	N/A	99 (Native)	Trunk Link to Switch0
PC2	Fa0	192.168.10.6	255.255.255.0	10	VLAN 10 Host
PC3	Fa0	192.168.10.7	255.255.255.0	10	VLAN 10 Host
PC4	Fa0	192.168.20.2	255.255.255.0	20	VLAN 20 Host
PC5	Fa0	192.168.20.3	255.255.255.0	20	VLAN 20 Host

3. Switch Configuration Procedures

Switch0 Setup

```
Switch0> enable
Switch0# configure terminal
! 1. Declare VLANs
Switch0(config)# vlan 10
Switch0(config-vlan)# name Users_VLAN10
Switch0(config-vlan)# vlan 99
Switch0(config-vlan)# name Management_Native99
```

```
! 2. Assign Access Ports
Switch0(config)# interface range FastEthernet0/2 - 3
Switch0(config-if-range)# switchport mode access
Switch0(config-if-range)# switchport access vlan 10

! 3. Configure Trunk Port
Switch0(config)# interface FastEthernet0/1
Switch0(config-if)# switchport mode trunk
Switch0(config-if)# switchport trunk native vlan 99

! 4. Management SVI
Switch0(config)# interface vlan 10
Switch0(config-if)# ip address 192.168.10.2 255.255.255.0
Switch0(config-if)# no shutdown
```

Switch1 Setup

```
Switch1> enable
Switch1# configure terminal
! 1. Declare VLANs
Switch1(config)# vlan 10
Switch1(config-vlan)# vlan 20
Switch1(config-vlan)# vlan 99

! 2. Assign Access Ports
Switch1(config)# interface range FastEthernet0/2 - 3
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 10

Switch1(config)# interface range FastEthernet0/4 - 5
Switch1(config-if-range)# switchport mode access
Switch1(config-if-range)# switchport access vlan 20

! 3. Configure Trunk Port
Switch1(config)# interface FastEthernet0/1
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# switchport trunk native vlan 99

! 4. Management SVIs
Switch1(config)# interface vlan 10
Switch1(config-if)# ip address 192.168.10.3 255.255.255.0
Switch1(config-if)# interface vlan 20
Switch1(config-if)# ip address 192.168.20.1 255.255.255.0
```